RICCARDO ZINI

Electronics & Automation Engineer

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Summary

Dedicated Electronics Engineering master's student with expertise in **embedded systems**, **analog and mixed-signal electronics**, and automated control. As Chief Engineer in a top-tier Formula Student team, I led an international group of 12, managing the full design-to-completion cycle of complex hardware and software under strict deadlines. While my background is rooted in high-performance motorsport, my ultimate goal is to apply my analytical problem-solving and technical skills to high-tech environments and fundamental scientific missions for humanity. I am deeply driven by the continuous progress of Engineering Physics and eager to bring my collaborative mindset to support groundbreaking research.

Key Achievements

Team Lead

Led a team of 12 engineers, enhancing cross-department collaboration.

Top Competition

Achieved top 5 finish in many Formula Student competitions.

Autonomous Vehicle

Successfully developed an autonomous vehicle prototype within only 8 months.

System Modelling

Completely modelled a EV from Powertrain to Dynamics, using Matlab/Simulink.

VCU System Management

Managed VCU system, enhancing performance by 25% during competitions.

Control System Development

Developed control systems for EV, from modelling to VCU RT implementation and testing on track

Experience

UniBo Motorsport - Formula SAE Team University of Bologna

Bologna, Italy
09/2025 - Present

Electronics & Power Electronics Engineer

- Developed firmware for embedded system for an EV Battery Management Systems (BMS).
- Engineered PCBs for BMS and high-connectivity vehicle systems.
- Rapidly integrated innovative custom SiC Inverters into vehicle architecture.
- Designed full-vehicle packaging to maximize spatial efficiency and racing performance.
- Implemented safety-critical hardware and software layers for Autonomous Driving systems.

UniBo Motorsport - Formula SAE Team University of Bologna

Bologna, Italy
09/2024 - 09/2025

Chief Engineer

- Led a 12-engineer team on EV Control Systems architecture and cross-functional integration.
- Managed technical PR and industrial partnerships, coordinating R&D and software development with corporate sponsors.
- Developed VCU firmware and real-time control algorithms using C and Embedded Linux.
- Architected the first Autonomous Driving prototype for the team, focusing on system infrastructure.
- Modeled EV Powertrain (Inverter, PMSM, Battery) in Simulink for dynamic performance optimization.
- Implemented ABS and Traction Control systems with independent 4-motor slip control.
- Managed technical B2B relations for motor control software and hardware development.

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Control System Engineer

- Developed MATLAB/Simulink control models for electric powertrain dynamics and vehicle state estimation.
- Implemented real-time firmware in C, enhancing VCU diagnostics and safety-critical fail-safe protocols.
- Optimized embedded software by refactoring legacy code, improving VCU processing efficiency and reliability.
- Contributed to high-performance vehicle setups for FSAustria, FSSpain, and FSATA international competitions.

Education

Alma Mater Studiorum - University of Bologna
Bachelor's Degree - Automation Engineering (2025)

Alma Mater Studiorum - University of Bologna
Masters's Degree - Electronics Engineering (Present)

Skills

Software Skills: NI LabVIEW • C • C++ • Git • Matlab • Simulink • STM32 • Virtuoso • Altium Designer • SPICE

Language & Soft Skills: **English** - **C1** • Strategic Leadership • Analytical Problem-solving • Adaptability

I consent to the processing of personal data present in the CV according to the General Data Protection Regulation (GDPR) 679/2016.